6 Activity Diagram (New Syntax)

The previous syntax used for activity diagrams encountered several limitations and maintainability issues. Recognizing these drawbacks, we have introduced a wholly revamped syntax and implementation that is not only user-friendly but also more stable.

6.0.1 Benefits of the New Syntax

- No Dependency on Graphviz: Just like with sequence diagrams, the new syntax eliminates the necessity for Graphviz installation, thereby simplifying the setup process.
- Ease of Maintenance: The intuitive nature of the new syntax means it is easier to manage and maintain your diagrams.

6.0.2 Transition to the New Syntax

While we will continue to support the old syntax to maintain compatibility, we highly encourage users to migrate to the new syntax to leverage the enhanced features and benefits it offers.

Make the shift today and experience a more streamlined and efficient diagramming process with the new activity diagram syntax.

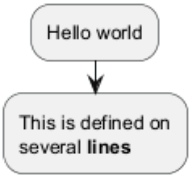
6.1 Simple action

Activities label starts with `:` and ends with `;`.

Text formatting can be done using creole wiki syntax.

They are implicitly linked in their definition order.

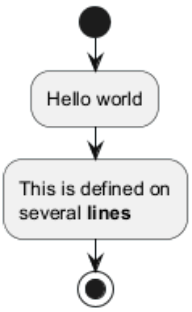
```
@startuml
:Hello world;
:This is defined on
several **lines**;
@enduml
```



6.2 Start/Stop/End

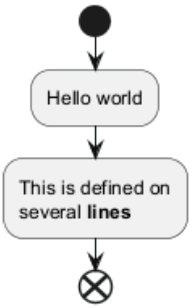
You can use `start` and `stop` keywords to denote the beginning and the end of a diagram.

```
@startuml
start
:Hello world;
:This is defined on
several **lines**;
stop
@enduml
```



You can also use the `end` keyword.

```
@startuml
start
:Hello world;
:This is defined on
several **lines**;
end
@enduml
```



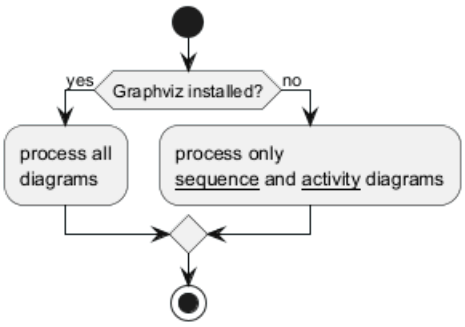
6.3 Conditional

You can use `if`, `then`, `else` and `endif` keywords to put tests in your diagram. Labels can be provided using parentheses.

The 3 syntaxes are possible:

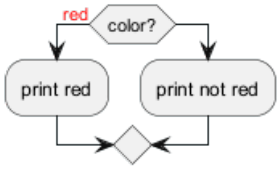
- `if (...) then (...) ... [else (...) ...] endif`

```
@startuml

start

if (Graphviz installed?) then (yes)
  :process all\ndiagrams;
else (no)
  :process only
  __sequence__ and __activity__ diagrams;
endif

stop

@enduml
```



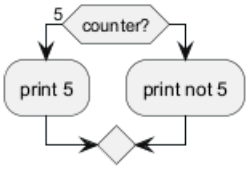
- `if (...) is (...) then ... [else (...) ...] endif`

```
@startuml
if (color?) is (<color:red>red) then
:print red;
else
:print not red;
endif
@enduml
```



- `if (...) equals (...) then ... [else (...) ...] endif`

```
@startuml
if (counter?) equals (5) then
:print 5;
else
:print not 5;
endif
@enduml
```



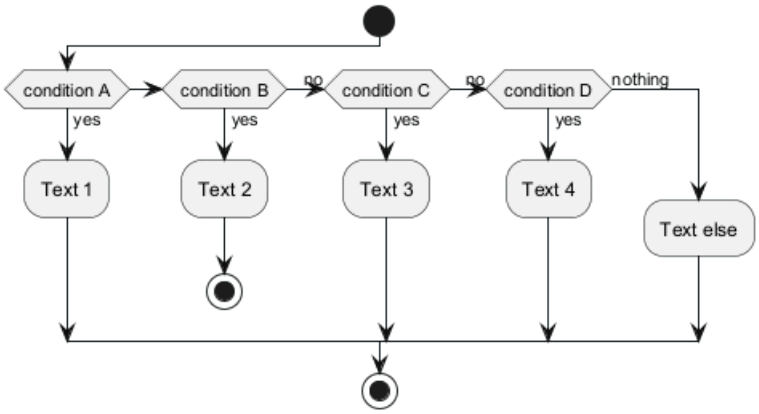
[Ref. QA-301]

6.3.1 Several tests (horizontal mode)

You can use the `elseif` keyword to have several tests *(by default, it is the horizontal mode)*:

```
@startuml
start
if (condition A) then (yes)
  :Text 1;
elseif (condition B) then (yes)
  :Text 2;
  stop
(no) elseif (condition C) then (yes)
  :Text 3;
(no) elseif (condition D) then (yes)
```

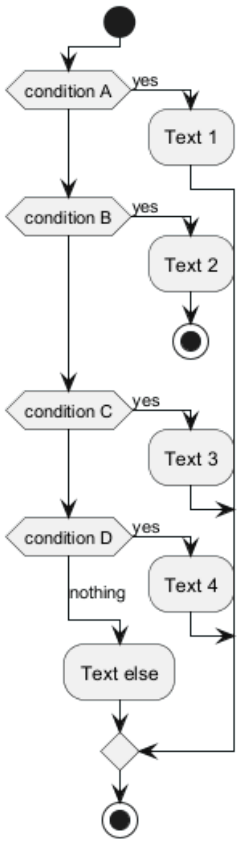
```
  :Text 4;
else (nothing)
  :Text else;
endif
stop
@enduml
```



6.3.2 Several tests (vertical mode)

You can use the command `!pragma useVerticalIf on` to have the tests in vertical mode:

```
@startuml
!pragma useVerticalIf on
start
if (condition A) then (yes)
  :Text 1;
elseif (condition B) then (yes)
  :Text 2;
  stop
elseif (condition C) then (yes)
  :Text 3;
elseif (condition D) then (yes)
  :Text 4;
else (nothing)
  :Text else;
endif
stop
@enduml
```



You can use the `-P` command-line option to specify the pragma:

```
java -jar plantuml.jar -PuseVerticalIf=on
```

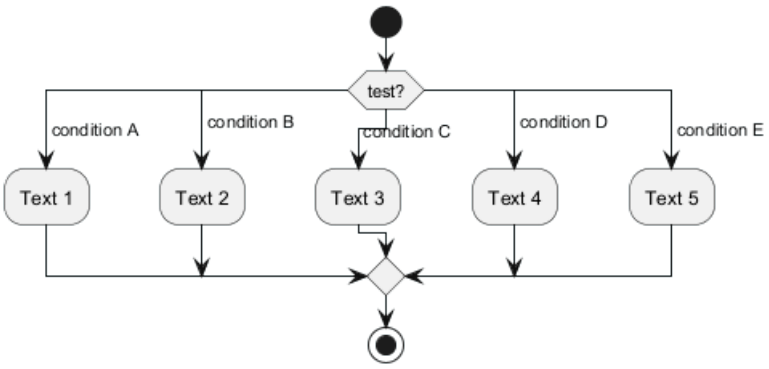
[Refs. QA-3931, GH-582]

6.4 Switch and case [switch, case, endswitch]

You can use `switch`, `case` and `endswitch` keywords to put switch in your diagram.

Labels can be provided using parentheses.

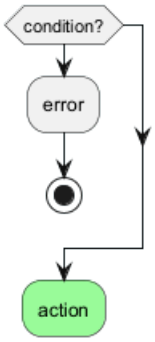
```
@startuml
start
switch (test?)
case ( condition A )
  :Text 1;
case ( condition B )
  :Text 2;
case ( condition C )
  :Text 3;
case ( condition D )
  :Text 4;
case ( condition E )
  :Text 5;
endswitch
stop
@enduml
```



6.5 Conditional with stop on an action [kill, detach]

You can stop action on a if loop.

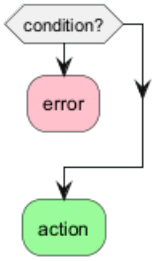
```
@startuml
if (condition?) then
  :error;
  stop
endif
#palegreen:action;
@enduml
```



But if you want to stop at the precise action, you can use the `kill` or `detach` keyword:

- `kill`

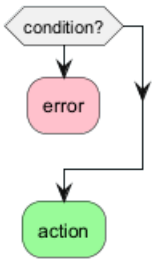
```
@startuml
if (condition?) then
  #pink:error;
  kill
endif
#palegreen:action;
@enduml
```



[Ref. QA-265]

- `detach`

```
@startuml
if (condition?) then
  #pink:error;
  detach
endif
#palegreen:action;
@enduml
```

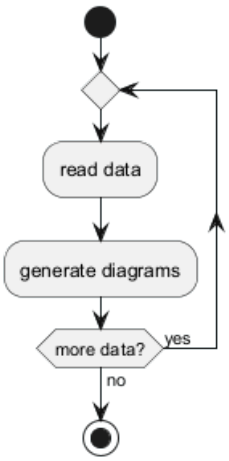


6.6 Repeat loop

6.6.1 Simple repeat loop

You can use `repeat` and `repeat while` keywords to have repeat loops.

```
@startuml

start

repeat
  :read data;
  :generate diagrams;
repeat while (more data?) is (yes) not (no)

stop

@enduml
```

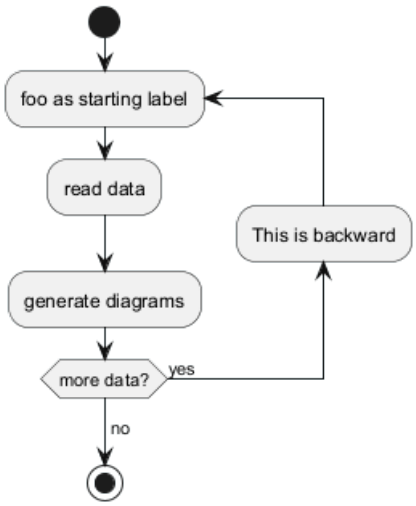


6.6.2 Repeat loop with repeat action and backward action

It is also possible to use a full action as `repeat` target and insert an action in the return path using the `backward` keyword.

```
@startuml

start
```

```
repeat :foo as starting label;
  :read data;
  :generate diagrams;
backward:This is backward;
repeat while (more data?) is (yes)
->no;

stop

@enduml
```

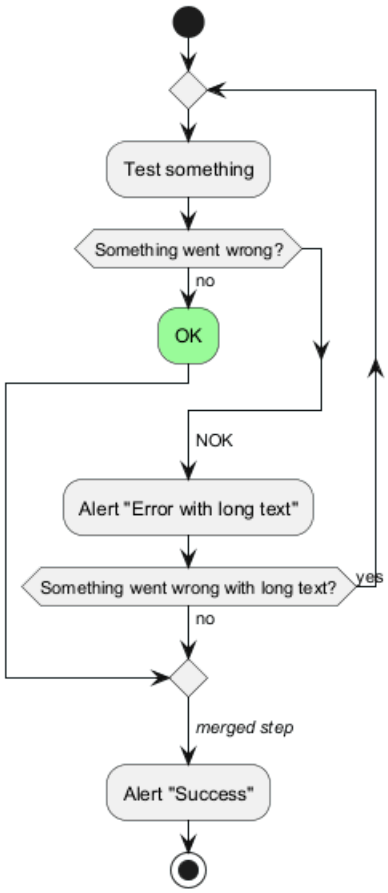


[Ref. QA-5826]

6.7 Break on a repeat loop [break]

You can use the `break` keyword after an action on a loop.

```
@startuml
start
repeat
  :Test something;
    if (Something went wrong?) then (no)
      #palegreen:OK;
      break
    endif
    ->NOK;
    :Alert "Error with long text";
repeat while (Something went wrong with long text?) is (yes) not (no)
->//merged step//;
:Alert "Success";
stop
@enduml
```



[Ref. QA-6105]

6.8 Goto and Label Processing [label, goto]

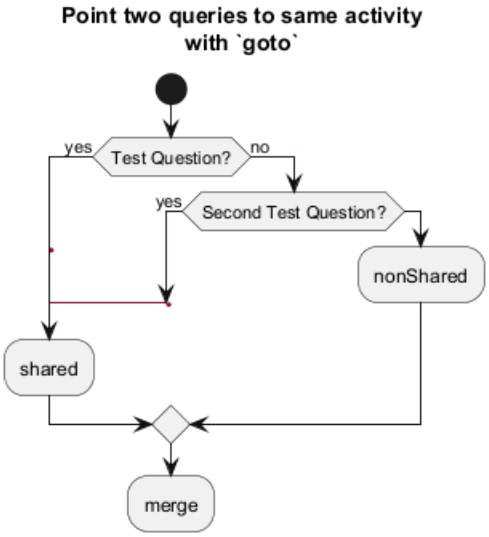
It is currently only experimental

You can use `label` and `goto` keywords to denote goto processing, with:

- `label <label_name>`
- `goto <label_name>`

```
@startuml
title Point two queries to same activity\nwith `goto`
start
if (Test Question?) then (yes)
'space label only for alignment
label sp_lab0
label sp_lab1
'real label
label lab
:shared;
else (no)
if (Second Test Question?) then (yes)
label sp_lab2
goto sp_lab1
else
:nonShared;
endif
endif
:merge;
```

```
@enduml
```



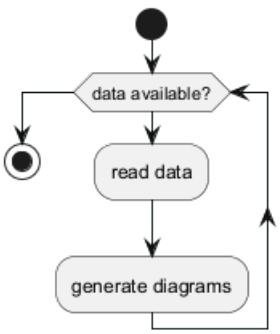
[Ref. QA-15026, QA-12526 and initially QA-1626]

6.9 While loop

6.9.1 Simple while loop

You can use `while` and `endwhile` keywords to have while loop.

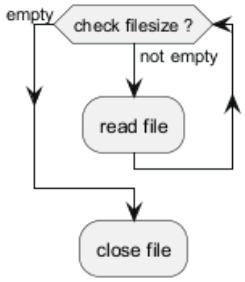
```
@startuml

start

while (data available?)
  :read data;
  :generate diagrams;
endwhile

stop

@enduml
```



It is possible to provide a label after the `endwhile` keyword, or using the `is` keyword.

```
@startuml
while (check filesize ?) is (not empty)
  :read file;
endwhile (empty)
:close file;
```

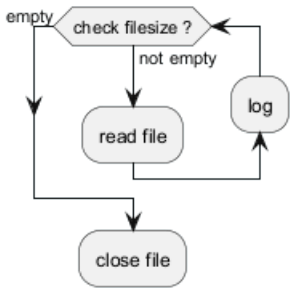
```
@enduml
```



6.9.2 While loop with backward action

It is also possible to insert an action in the return path using the `backward` keyword.

```
@startuml
while (check filesize ?) is (not empty)
  :read file;
  backward:log;
endwhile (empty)
:close file;
@enduml
```

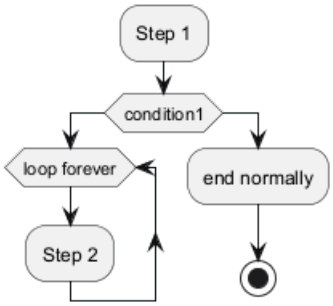


[Ref. QA-11144]

6.9.3 Infinite while loop

If you are using `detach` to form an infinite while loop, then you will want to also hide the partial arrow that results using `-[hidden]->`

```
@startuml
:Step 1;
if (condition1) then
  while (loop forever)
   :Step 2;
  endwhile
  -[hidden]->
  detach
else
  :end normally;
  stop
endif
@enduml
```

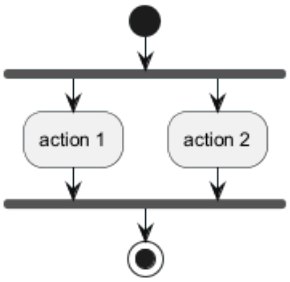


6.10 Parallel processing [fork, fork again, end fork, end merge]

You can use `fork`, `fork again` and `end fork` or `end merge` keywords to denote parallel processing.

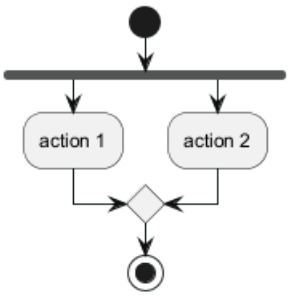
6.10.1 Simple `fork`

```
@startuml
start
fork
  :action 1;
fork again
  :action 2;
end fork
stop
@enduml
```



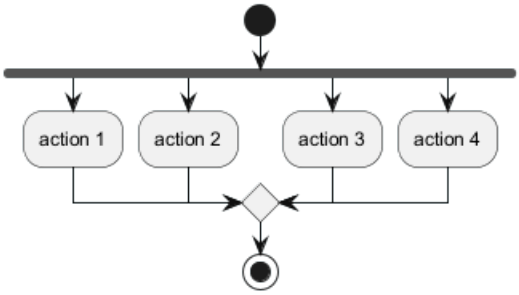
6.10.2 `fork` with end merge

```
@startuml
start
fork
  :action 1;
fork again
  :action 2;
end merge
stop
@enduml
```

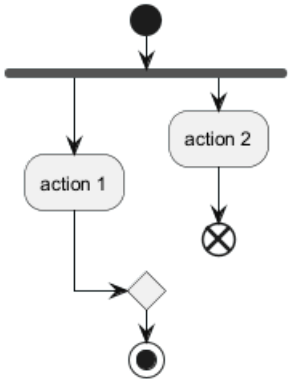


[Ref. QA-5320]

```
@startuml
start
fork
  :action 1;
fork again
  :action 2;
fork again
  :action 3;
fork again
  :action 4;
end merge
stop
@enduml
```




```
@startuml
start
fork
  :action 1;
fork again
  :action 2;
  end
end merge
stop
@enduml
```

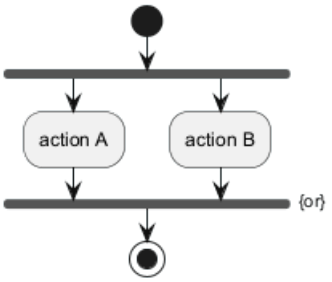


[Ref. QA-13731]

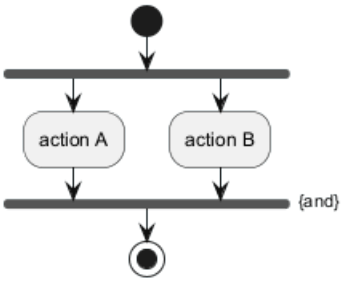
6.10.3 Label on `end fork` (or UML joinspec):

```
@startuml
start
fork
  :action A;
fork again
```

```
  :action B;
end fork {or}
stop
@enduml
```



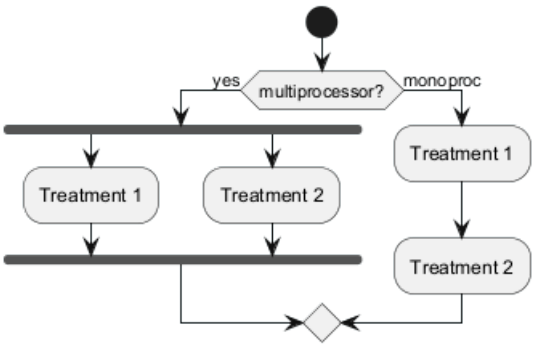

```
@startuml
start
fork
  :action A;
fork again
  :action B;
end fork {and}
stop
@enduml
```



[Ref. QA-5346]

6.10.4 Other example

```
@startuml

start

if (multiprocessor?) then (yes)
  fork
    :Treatment 1;
  fork again
    :Treatment 2;
  end fork
else (monoproc)
  :Treatment 1;
  :Treatment 2;
endif

@enduml
```

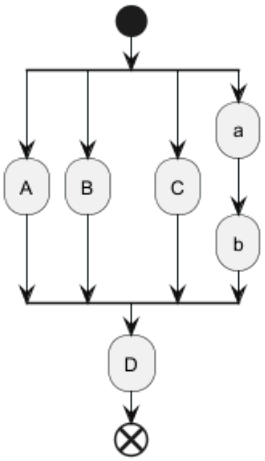


6.11 Split processing

6.11.1 Split

You can use `split`, `split again` and `end split` keywords to denote split processing.

```
@startuml
start
split
   :A;
split again
   :B;
split again
   :C;
split again
   :a;
   :b;
end split
:D;
end
@enduml
```

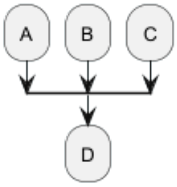


6.11.2 Input split (multi-start)

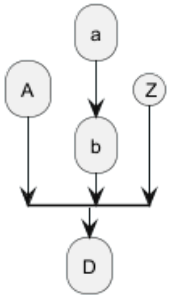
You can use `hidden` arrows to make an input split (multi-start):

```
@startuml
split
   -[hidden]->
   :A;
split again
   -[hidden]->
```

```
   :B;
split again
   -[hidden]->
   :C;
end split
:D;
@enduml
```




```
@startuml
split
   -[hidden]->
   :A;
split again
   -[hidden]->
   :a;
   :b;
split again
   -[hidden]->
   (Z)
end split
:D;
@enduml
```



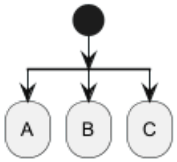
[Ref. QA-8662]

6.11.3 Output split (multi-end)

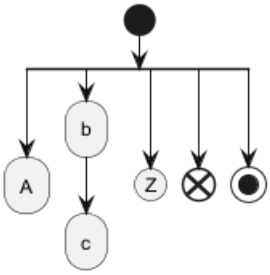
You can use `kill` or `detach` to make an output split (multi-end):

```
@startuml
start
split
   :A;
   kill
split again
   :B;
   detach
split again
   :C;
   kill
```

```
end split
@enduml
```




```
@startuml
start
split
   :A;
   kill
split again
   :b;
   :c;
   detach
split again
   (Z)
   detach
split again
   end
split again
   stop
end split
@enduml
```

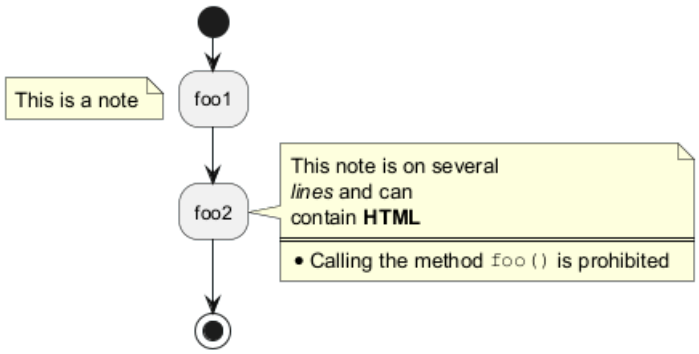


6.12 Notes

Text formatting can be done using creole wiki syntax.

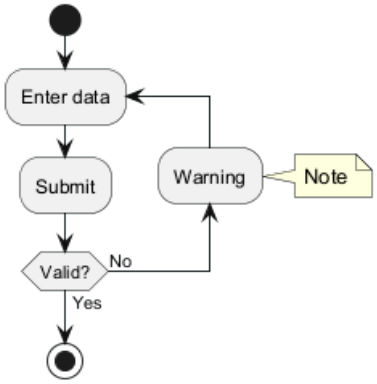
A note can be floating, using `floating` keyword.

```
@startuml

start
:foo1;
floating note left: This is a note
:foo2;
note right
  This note is on several
  //lines// and can
  contain <b>HTML</b>
  ====
  * Calling the method ""foo()"" is prohibited
end note
stop

@enduml
```



You can add note on backward activity:

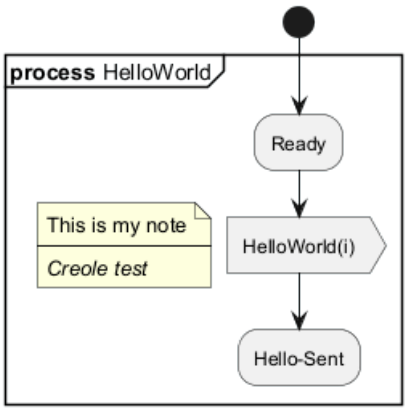
```
@startuml
start
repeat :Enter data;
:Submit;
backward :Warning;
note right: Note
repeat while (Valid?) is (No) not (Yes)
stop
@enduml
```



[Ref. QA-11788]

You can add note on partition activity:

```
@startuml
start
partition "**process** HelloWorld" {
    note
        This is my note
        ----
        //Creole test//
    end note
    :Ready;
    :HelloWorld(i)>
    :Hello-Sent;
}
@enduml
```

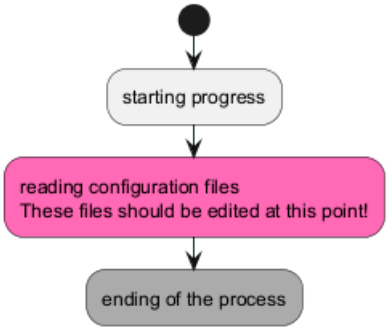


[Ref. QA-2398]

6.13 Colors

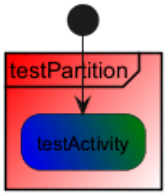
You can specify a color for some activities.

```
@startuml

start
:starting progress;
#HotPink:reading configuration files
These files should be edited at this point!;
#AAAAAA:ending of the process;

@enduml
```



You can also use gradient color.

```
@startuml
start
partition #red/white testPartition {
        #blue\green:testActivity;
}
@enduml
```

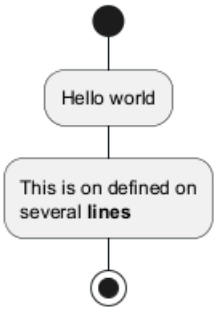


[Ref. QA-4906]

6.14 Lines without arrows

You can use `skinparam ArrowHeadColor none` in order to connect activities using lines only, without arrows.

```
@startuml
skinparam ArrowHeadColor none
start
:Hello world;
:This is on defined on
several **lines**;
stop
@enduml
```




```
@startuml
skinparam ArrowHeadColor none
start
repeat :Enter data;
:Submit;
backward :Warning;
repeat while (Valid?) is (No) not (Yes)
stop
@enduml
```

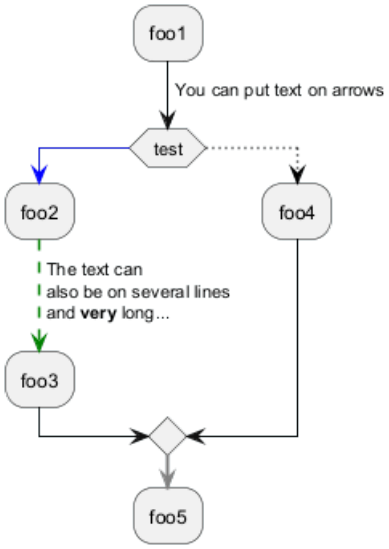
6.15 Arrows

Using the `->` notation, you can add texts to arrow, and change their color.

It's also possible to have dotted, dashed, bold or hidden arrows.

```
@startuml
:foo1;
-> You can put text on arrows;
if (test) then
  -[#blue]->
  :foo2;
  -[#green,dashed]-> The text can
```

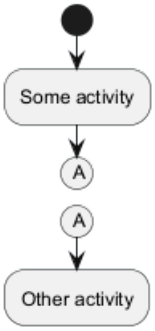
```
  also be on several lines
  and **very** long...;
  :foo3;
else
  -[#black,dotted]->
  :foo4;
endif
-[#gray,bold]->
:foo5;
@enduml
```



6.16 Connector

You can use parentheses to denote connector.

```
@startuml
start
:Some activity;
(A)
detach
(A)
:Other activity;
@enduml
```

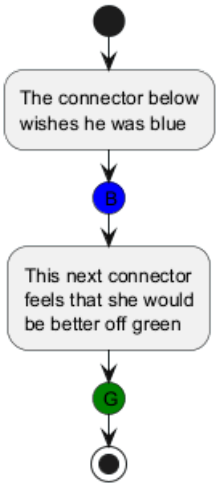


6.17 Color on connector

You can add color on connector.

```
@startuml
```

```
start
:The connector below
wishes he was blue;
#blue:(B)
:This next connector
feels that she would
be better off green;
#green:(G)
stop
@enduml
```



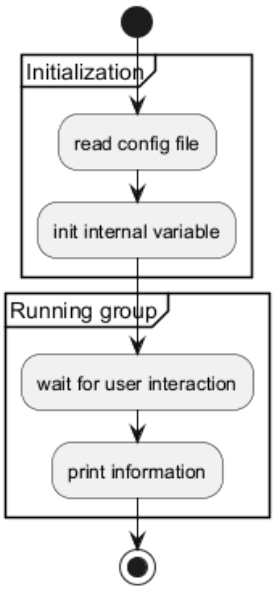
[Ref. QA-10077]

6.18 Grouping or partition

6.18.1 Group

You can group activity together by defining group:

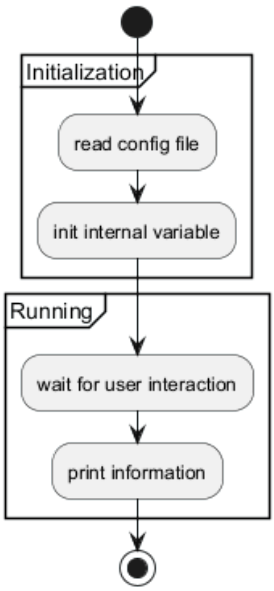
```
@startuml
start
group Initialization
    :read config file;
    :init internal variable;
end group
group Running group
    :wait for user interaction;
    :print information;
end group

stop
@enduml
```



6.18.2 Partition

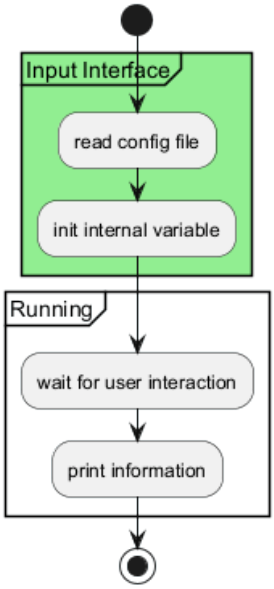
You can group activity together by defining partition:

```
@startuml
start
partition Initialization {
    :read config file;
    :init internal variable;
}
partition Running {
    :wait for user interaction;
    :print information;
}

stop
@enduml
```



It's also possible to change partition color:

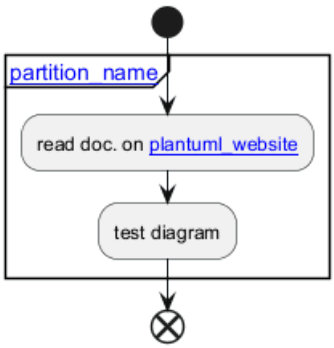
```
@startuml
start
partition #lightGreen "Input Interface" {
    :read config file;
    :init internal variable;
}
partition Running {
    :wait for user interaction;
    :print information;
}
stop
@enduml
```



[Ref. QA-2793]

It's also possible to add link to partition:

```
@startuml
start
partition "[[http://plantuml.com partition_name]]" {
    :read doc. on [[http://plantuml.com plantuml_website]];
    :test diagram;
}
end
@enduml
```



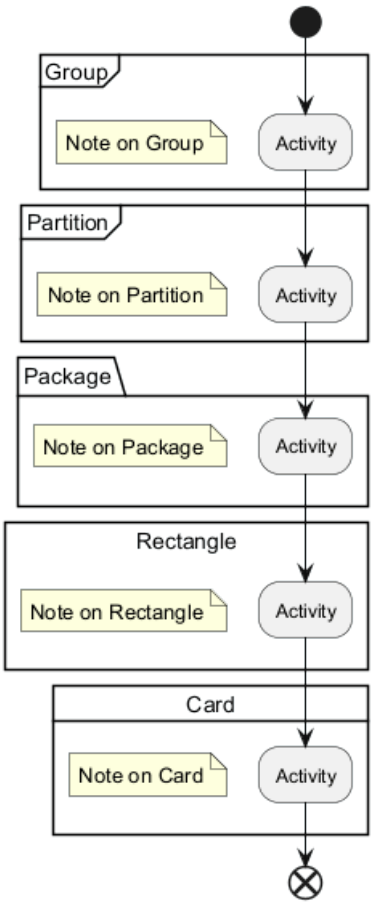
[Ref. QA-542]

6.18.3 Group, Partition, Package, Rectangle or Card

You can group activity together by defining:

- group;
- partition;
- package;
- rectangle;
- card.

```
@startuml
start
group Group
  :Activity;
end group
floating note: Note on Group

partition Partition {
  :Activity;
}
floating note: Note on Partition

package Package {
  :Activity;
}
floating note: Note on Package

rectangle Rectangle {
  :Activity;
}
floating note: Note on Rectangle

card Card {
  :Activity;
}
floating note: Note on Card
end
@enduml
```

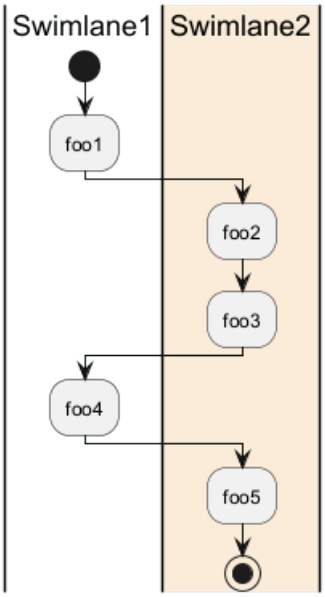


6.19 Swimlanes

Using pipe |, you can define swimlanes.

It's also possible to change swimlanes color.

```
@startuml
|Swimlane1|
start
:foo1;
|#AntiqueWhite|Swimlane2|
:foo2;
:foo3;
|Swimlane1|
:foo4;
|Swimlane2|
:foo5;
stop
@enduml
```



You can add `if` conditional or `repeat` or `while` loop within swimlanes.

```
@startuml
|#pink|Actor_For_red|
start
if (color?) is (red) then
#pink:**action red**;
:foo1;
else (not red)
|#lightgray|Actor_For_no_red|
#lightgray:**action not red**;
:foo2;
endif
|Next_Actor|
#lightblue:foo3;
:foo4;
|Final_Actor|
#palegreen:foo5;
stop
@enduml
```